HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and executed test protocols for electronic hardware and embedded software systems, ensuring compliance with technical requirements for medical devices.
- Authored detailed test plans and protocols, including fixture specifications and configuration details, to ensure full requirement traceability and reproducible test methods.
- Investigated and documented test failures and nonconformities, performing root cause analysis to identify underlying issues in hardware-software integrations.
- Collaborated with cross-functional R&D; teams to define hardware and software verification and validation scope, contributing to estimates and identifying regulatory risks.
- Assisted in ensuring verification and validation activities adhered to relevant medical device regulations and standards, including IEC/EN60601 and IEC/EN62304.
- Wrote and maintained scripts in **Python** for automating test execution and data analysis, reducing manual testing effort by approximately **30%**.
- Containerized test environments using **Docker** and managed deployments on **AWS EC2** instances, improving test setup consistency and reducing deployment times by **45%**.

SKILLS

Languages – Python, C, C++, JavaScript, SQL

Testing & QA – Verification, Validation, Test Planning, Test Execution, Protocol Development, Requirements Traceability, Defect Reporting, Root Cause Analysis

Embedded Systems – Embedded Software, Microcontrollers, Hardware-Software Systems, Schematic Review, Component-Level Testing

Tools & Technologies – Git, JIRA, Microsoft Word, Microsoft Excel, Microsoft PowerPoint, Copilot, Docker, AWS EC2, Oscilloscopes, Multimeters, Power Supplies

Medical Device Standards – IEC/EN60601, IEC/EN62304, FDA Guidance Documents

PROJECTS

Embedded Software Test Automation Framework

- Developed an automated test framework using **Python** to execute test cases against embedded firmware for a medical device controller, increasing test coverage by **20%**.
- Integrated the framework with hardware-in-the-loop (HIL) test equipment, including oscilloscopes and signal generators, to simulate real-world operating conditions.
- Authored comprehensive test reports and managed defect tracking within **JIRA**, providing objective evidence of system compliance to quality assurance.

Hardware-Software Integration Testing Suite

- Designed and implemented test protocols for validating the integration between custom electronic hardware and its embedded software, focusing on microcontrollers and sensor interfaces.
- Executed component-level and system-level tests, isolating faults and documenting findings using **Microsoft Excel** and **Confluence** for the engineering team.
- Collaborated with hardware and software engineers to refine test methodologies and improve the overall reliability of interconnected system components.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)